Fiscal Year:	FY 2014	**Task Last Updated:**	FY 05/12/2021
PI Name:	Furst, Eric Ph.D.		
Project Title:	Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:	**TechPort:**		No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19716	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	03/01/2010	**End Date:**	04/19/2018
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	InSPACE ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AD21G ; NNX10AE44G ; NNX07AD02G (InSPACE2)		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	InSPACE is a microgravity fluid physics experiment that was performed on the International Space Station (ISS). InSPACE began in December 1997 and concluded with the third series of follow-on InSPACE experiments aboard the ISS in November 2013. [Ed. note April 2021: added to Task Book when received information. Note that this investigator's (Dr. Eric Furst) original grant was InSPACE-2: Structural and Rheological Transitions of Field-Responsive Fluids in Microgravity, grant NNX07AD02G; period of performance was 1/8/2007-1/07/2010. Project continued with Dr. Furst as PI 3/1/2010-1/29/2016 as Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3), grant NNX10AE44G; 10/20/2015-4/19/2018, grant NNX16AD21G with same title, Investigating the Structure of Paramagnetic Aggregates from Colloidal Ellipsoids-3 (InSPACE 3)] InSPACE-2 was a continuation of the InSPACE investigation begun on ISS Expedition 6, providing new and improved samples for operation in the Microgravity Science Glovebox (MSG). The use of external fields to control the microstructure of colloidal suspensions has long been recognized as a powerful means for tailoring the mechanical, optical, and electronic properties of materials. Magnetorheological (MR) suspensions, in particular, provide a striking example. These normally stable fluids undergo a dynamic transition to a solid within milliseconds after the application of an external magnetic field. They are also important models for developing methods of bottom-up fabrication of micro- and nano-structured materials and devices using field-directed self-assembly. Investigating the Structure of Paramagnetic Aggregates from Colloidal Emulsions-2 (InSPACE-2) experiments focused on the structure of magnetically-polarizable particle suspensions over long times in steady (DC) and pulsed magnetic fields. In pulsed fields especially, the long-time kinetics of the suspension micro-structural coarsening provided new and important results. The InSPACE-2 experiments were the first to yield information on the full, three-dimensional aggregation process over timescales much longer than those that are accessible on the ground, which are limited by catastrophic sedimentation. Furthermore, the experiments identified a novel and intriguing dynamic instability in which the suspension microstructures were observed to buckle at specific field frequencies and field strengths. For InSPACE-3, we will investigate the effect of particle shape on the micro-structural evolution of MR suspensions. Recent ground-based experiments demonstrate a startling effect that		

	particle shape has on the interactions of dipolar chains and the resulting suspension microstructure. Specifically, we will investigate suspensions of paramagnetic ellipsoid-shaped particles. In combination with the results of InSPACE-2, we hypothesize that particle shape will dramatically alter the aggregation kinetics, microstructures, and microscopic mechanics. This will potentially lead to the ability to engineer enhanced properties of these suspensions, including suppression of the lateral aggregation in magnetorheological fluid-based electromechanical devices or the ability to create new colloidal materials through field-directed self-assembly.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Space Applications: The quick phase-shifting property of magnetic colloidal fluids makes them potentially useful for devices such as optical interfaces, active noise, and vibration dampers inside spacecrafts. Future uses could include robotics, energy-transfer devices such as clutches, and other control systems. Earth Applications: The work has application to directed self-assembly of crystalline structures from particles that may eventually allow creation of new nano-materials fabricated from nanoparticle building blocks, with potential applications in medicine, energy storage, chemical separations, and catalysis. Magnetic colloidal fluid technology is currently being used for shock absorbers in racecars and sport cars. This technology is expanding to include making large-scale building foundation stabilizers for areas prone to earthquakes.
Task Progress:	[Note (May 2021): compiled from PI's March 2014 Progress report for grant NNX10AE44G] Achievements in 2013 During the past year, we have (1) completed the InSPACE-3 MSG experiments, (2) performed preliminary analysis of the microgravity data, (3) performed ground-based experiments to provide context for the microgravity research, and (4) published and presented research from InSPACE-2 and InSPACE-3. In addition to research related to directed self-assembly in toggled fields, the theoretical tools we developed and used enabled studies of several related areas, including the response of colloids to electric fields, colloidal hydrodynamic interactions (directly applicable to biophysical mechanics experiments), and suspension rheology. The InSPACE project has had excellent visibility and impact. In addition to high-profile peer reviewed papers in Soft Matter, Physical Review Letters, and Proceedings of the National Academy of Sciences, InSPACE research was presented at several keynote and invited lectures, including the 2013 Soft Matter Lectureship at the International Soft Matter Conference in Rome, Italy, the American Astronautical Society meeting in Denver, Colorado, and the 2014 American Physical Society annual meeting. Furst was guest editor of the Soft Matter themed issue on Directed Self-Assembly. The accompanying editorial highlighted InSPACE work on toggled field assembly. The InSPACE project has received press coverage and Furst has participated in several outreach activities, including Space Station Live and Science@NASA. A list of online stories is provided below. In addition to the deliverable outcomes, the InSPACE project has had significant impact in education and human resource development. Former postdocs James Swan started as an assistant professor at MIT in September, 2013, with a primary appointment in Chemical Engineering, and Paula Vasquez started as a faculty member in the Department of Mathematics at the University of South Carolina. Two high school students involved in this work received recognitions: Martin Kurian was awarded prizes at the New Castle County and Delaware Valley Science Fairs and received the Dietrich W. Botstiber Foundation scholarship based on his research. Johnathan Stauffer presented his InSPACE-related research at the Cecil County Capstone gallery walk and matriculated as a freshman at CalTech in October 2013. Press and outreach 1. Space Station Live with Pat Ryan, February 6, 2014. http://youtu.be/ 2. Genius Materials on the ISS, Science@NASA, November 27, 2013. http://science.nasa.gov/ 3. ISS Research top station results, November 4, 2013. 4. Top Space Station Research Awards of the Year NASA, July 16, 2013. http://www.nasa.gov/ 5. Smart fluids could allow chips to assemble themselves R&D Magazine, September 19, 2012. 6. Directed Self-Assembly Accomplished with Magnets IEEE Spectrum September 20, 2012. http://spectrum.ieee.org/ 7. Out of this world: UD professor reports smart fluids research in scientific journal. UDaily Sept. 18, 2012. http://www.udel.edu/ 8. Out-of-this-world nanoscience. University of Delaware Research Vol. 3 No. 2, 2012. http://www.udel.edu/
Bibliography Type:	Description: (Last Updated: 01/27/2026)
Articles in Peer-reviewed Journals	Swan JW, Vasquez PA, Whitson PA, Fincke EM, Wakata K, Magnus SH, De Winne F, Barratt MR, Agui JH, Green RD, Hall NR, Bohman DY, Bunnell CT, Gast AP, Furst EM. "Multi-scale kinetics of a field-directed colloidal phase transition." Proc Natl Acad Sci U S A. 2012 Oct 2;109(40):16023-8. https://doi.org/10.1073/pnas.1206915109 ; PMID: 22988079; PMCID: PMC3479562 , Oct-2012
Articles in Peer-reviewed Journals	Swan JW, Shindel MM, Furst EM. "Measuring thermal rupture force distributions from an ensemble of trajectories." Phys Rev Lett. 2012 Nov 9;109(19):198302. https://doi.org/10.1103/PhysRevLett.109.198302 ; PMID: 23215431 , Nov-2012
Articles in Peer-reviewed Journals	Swan JW, Furst EM. "A simpler expression for Henry's function describing the electrophoretic mobility of spherical colloids." J Colloid Interface Sci. 2012 Dec 15;388(1):92-4. https://doi.org/10.1016/j.jcis.2012.08.026 ; PMID: 22995247 , Dec-2012
Articles in Peer-reviewed Journals	Swan JW, Furst EM. "Nonequilibrium distributions and hydrodynamic coupling distort the measurement of nanoscale forces near interfaces." Biophys J. 2013 Feb 19;104(4):863-72. https://doi.org/10.1016/j.bpj.2012.12.014 ; PMID: 23442965; PMCID: PMC3576541 , Feb-2013
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Journal/Magazine covers	Swan JW, Bauer JL, Liu Y, Furst EM. "Cover in journal Soft Matter for article, 'Directed colloidal self-assembly in toggled magnetic fields.' " Soft Matter. 2014 Feb 28;10(8):1102-9. https://doi.org/10.1039/c3sm52663a ; https://pubs.rsc.org/en/content/articlelanding/2014/sm/c4sm90011a#!divAbstract , Feb-2014